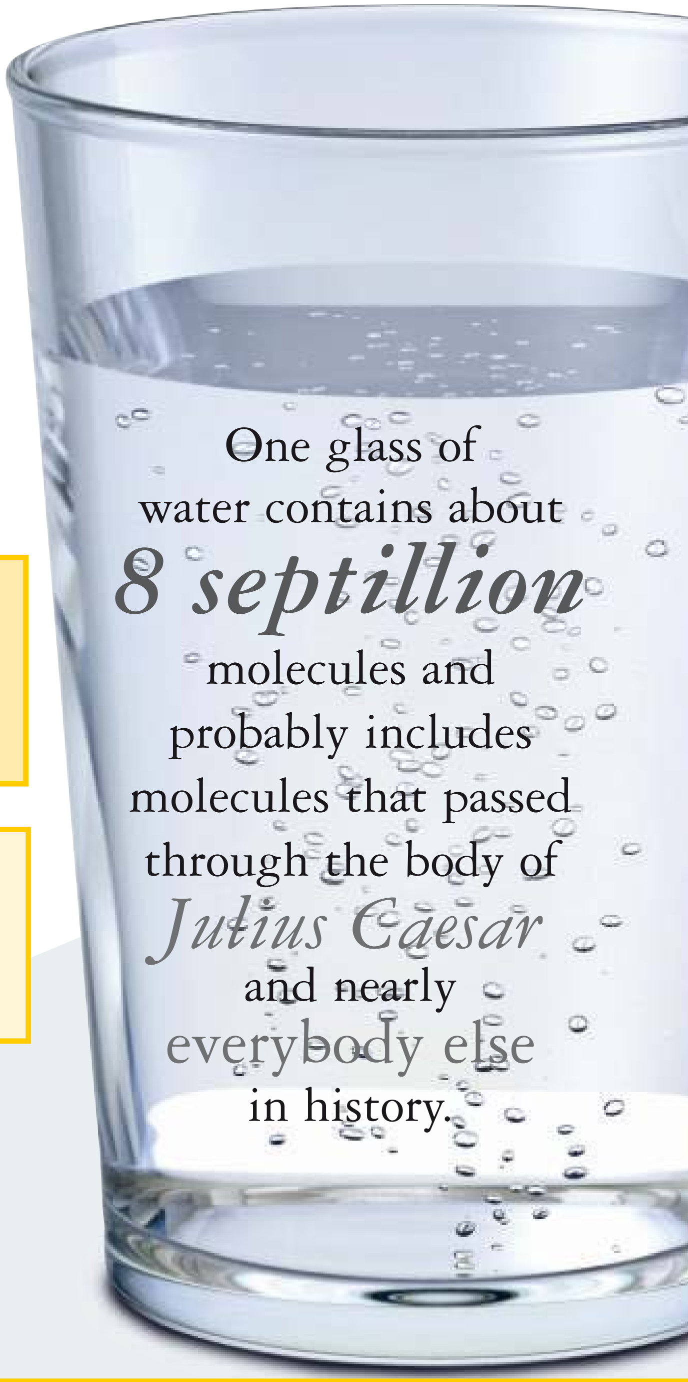
BIG NUMBERS

A thousand has three zeros, a million has six. Each time you add three more zeros, you reach a number with a new name.

- thousand
- million
- billion
- trillion
- quadrillion
- quintillion
- sexillion

How many drops of water make an ocean? How many atoms are there in your body? How many grains of sand would fill the universe? Some numbers are so big we can't imagine them or even write them down. Mathematicians cope with these whoppers by using "*powers*".



WHAT ARE POWERS?

A power is a tiny number written just next to another number, like this: 4^2

It means "4 to the power of 2".

> **The power tells you how many times to multiply the main number by itself. 4^2 means multiply two fours together: 4×4, which is 16. And 4^3 means $4 \times 4 \times 4$, which is 64.**

Power crazy

Powers are handy because they make it easy to write down numbers that would otherwise be much too long. Take the number 9^{9^9}, for instance, which means 9 to the power of 9 to the power of 9, or $9^{387,420,489}$. If you wrote this in full, you'd need 369 million digits and a piece of paper 800 km (500 miles) long.

1 GOOGOL = 10,000,000,000,000,000,000,000,000,000,000,000,000,000,000,000,000,00